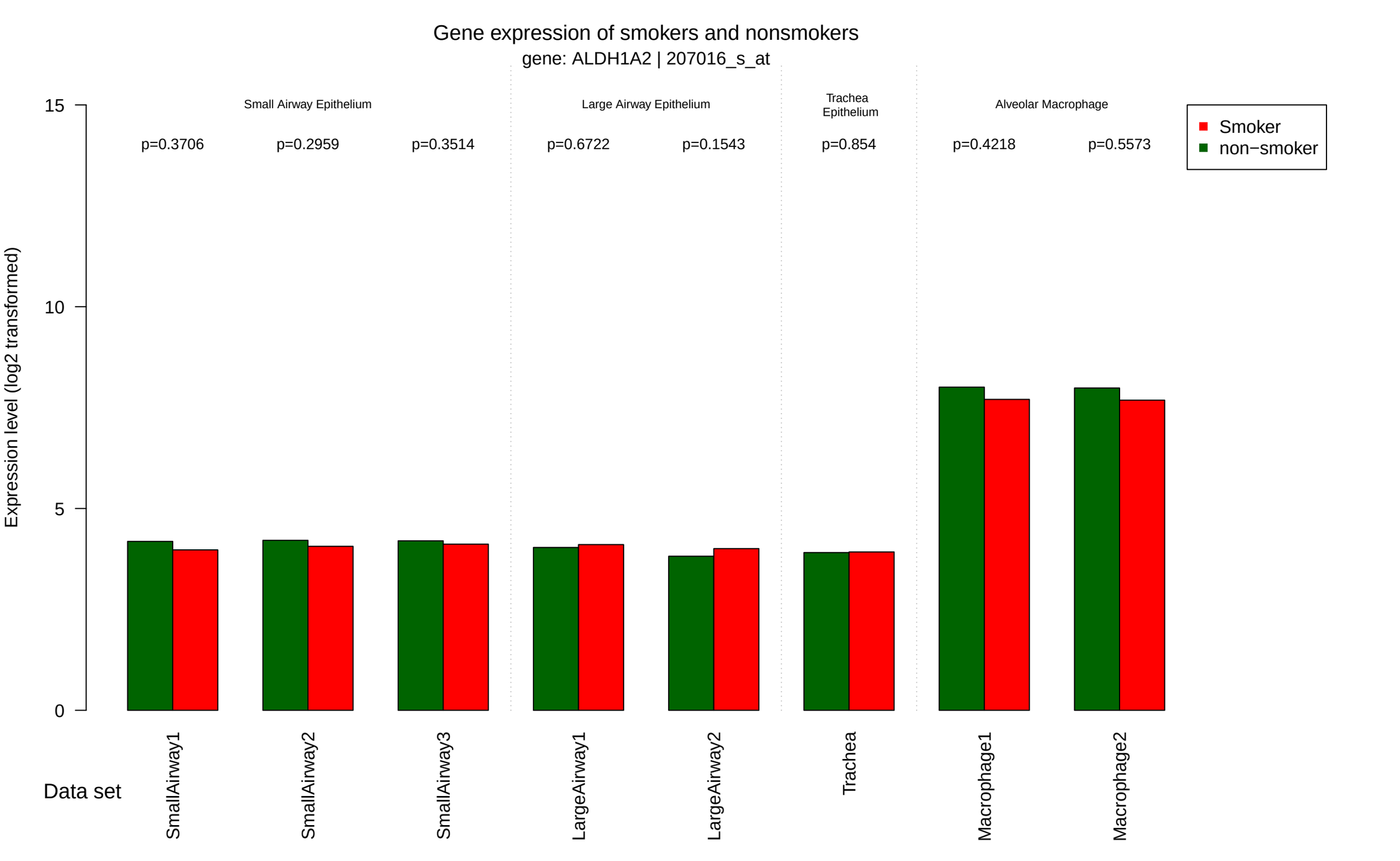
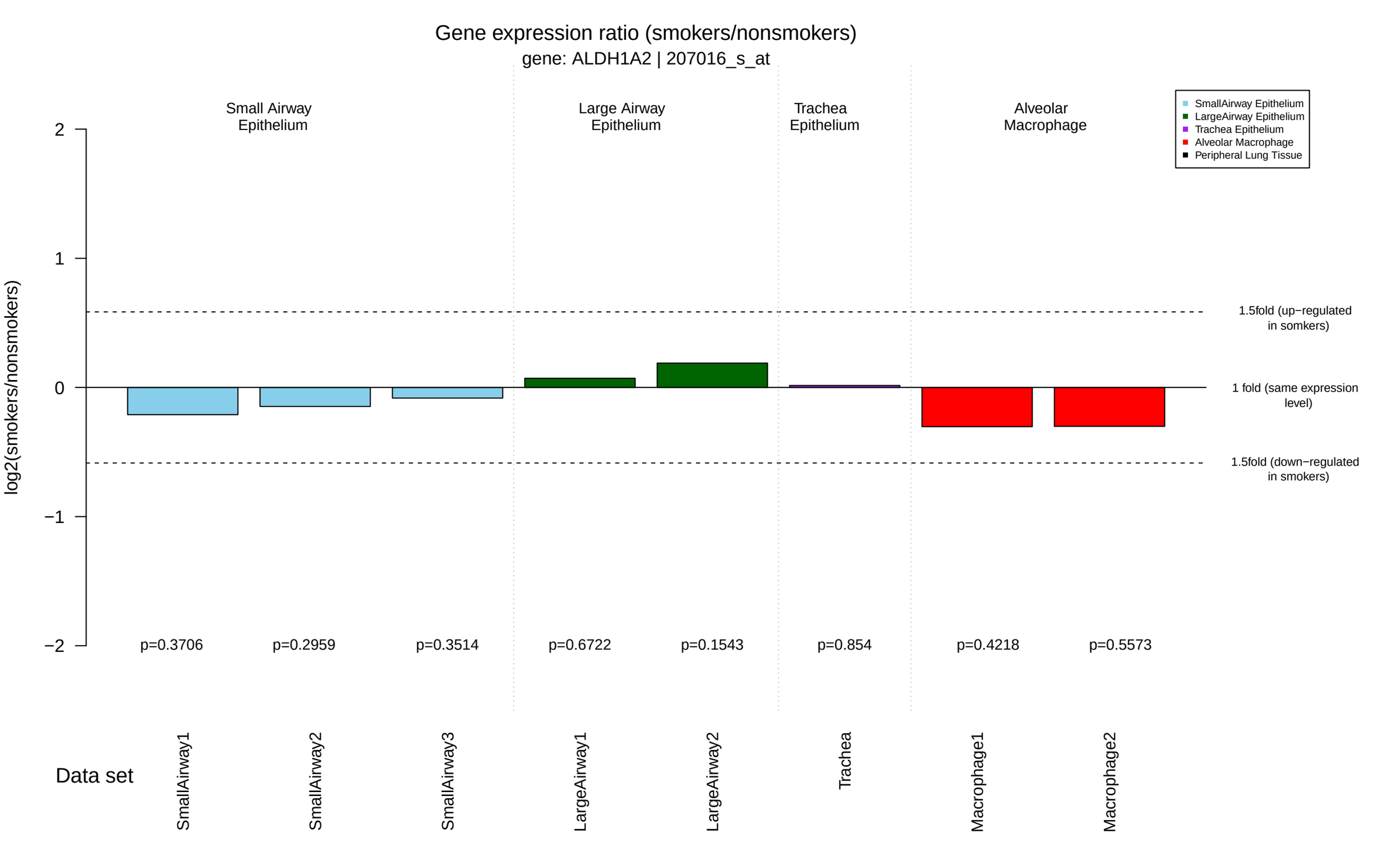
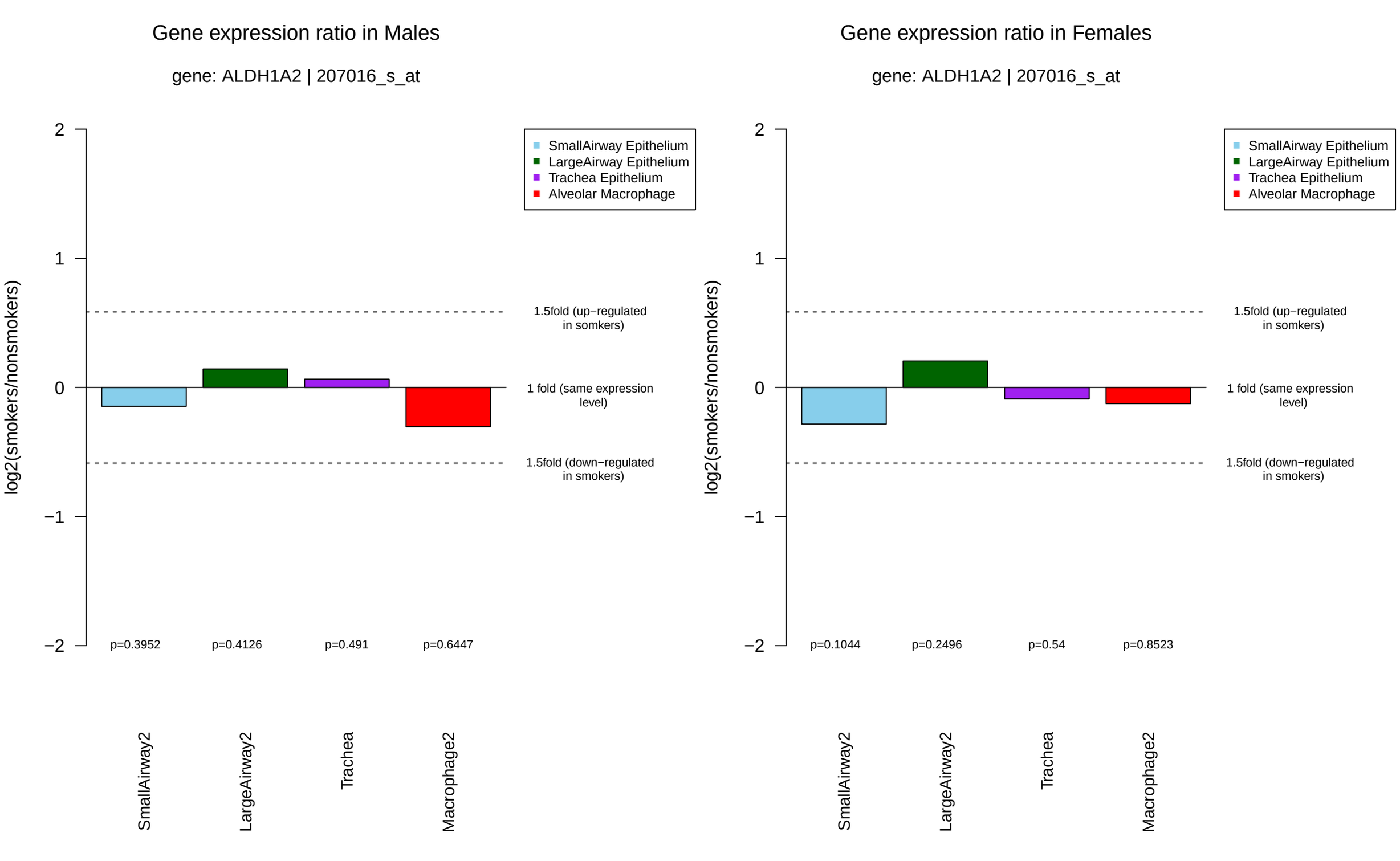
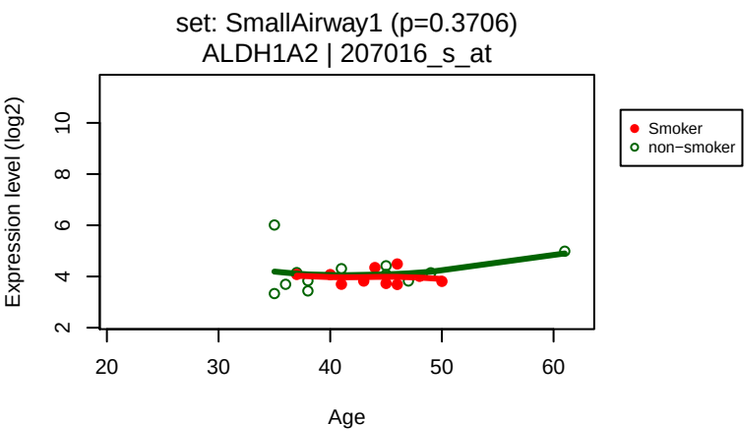
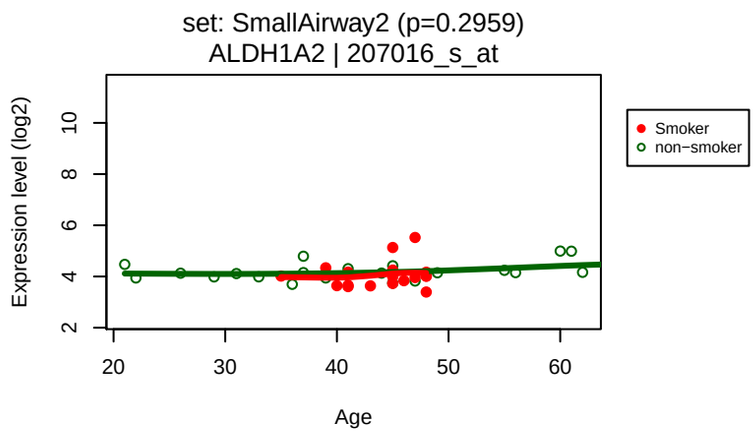
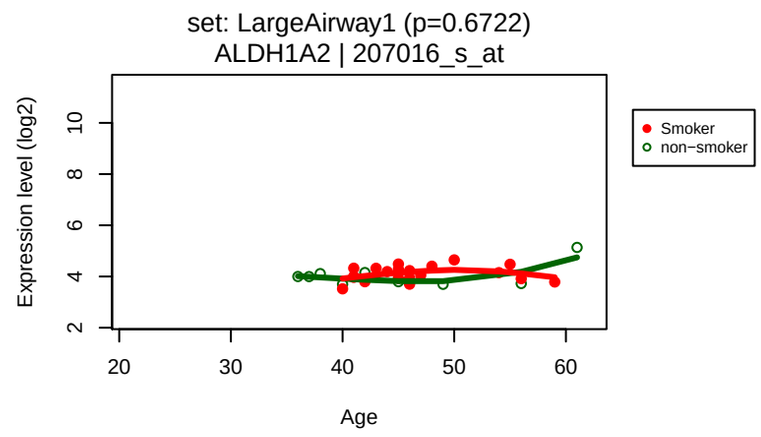

Gene expression of smokers and nonsmokers
gene: ALDH1A2 | 207016_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3706
p=0.2959
p=0.3514
p=0.6722
p=0.1543
p=0.854
p=0.4218
p=0.5573
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

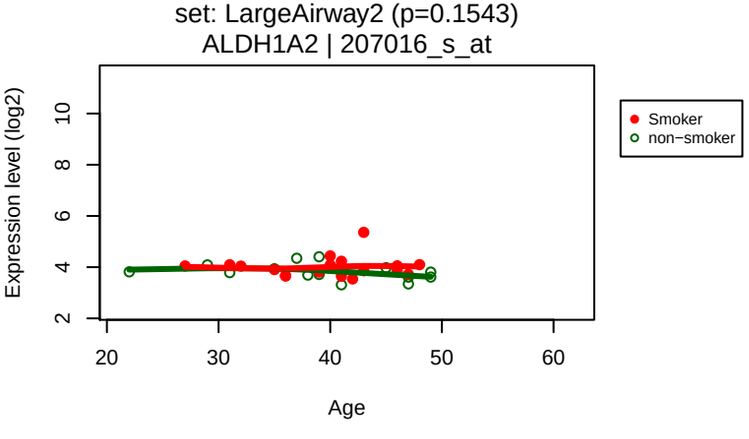

Gene expression ratio (smokers/nonsmokers)
gene: ALDH1A2 | 207016_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3706
p=0.2959
p=0.3514
p=0.6722
p=0.1543
p=0.854
p=0.4218
p=0.5573
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

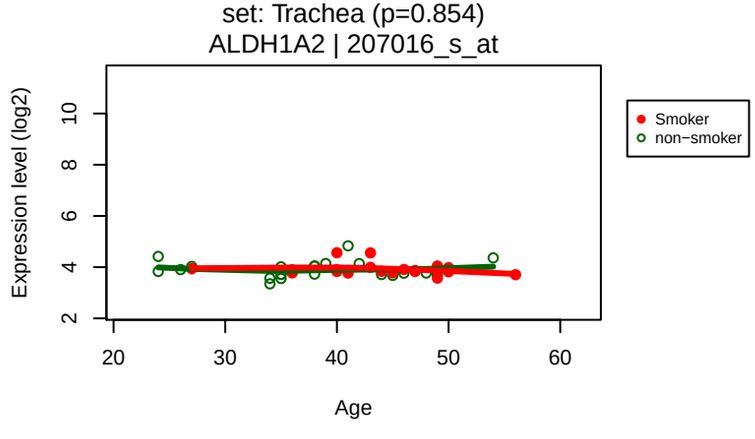

Gene expression ratio in Males
gene: ALDH1A2 | 207016_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3952
p=0.4126
p=0.491
p=0.6447
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALDH1A2 | 207016_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1044
p=0.2496
p=0.54
p=0.8523
SmallAirway2
LargeAirway2
Trachea
Macrophage2

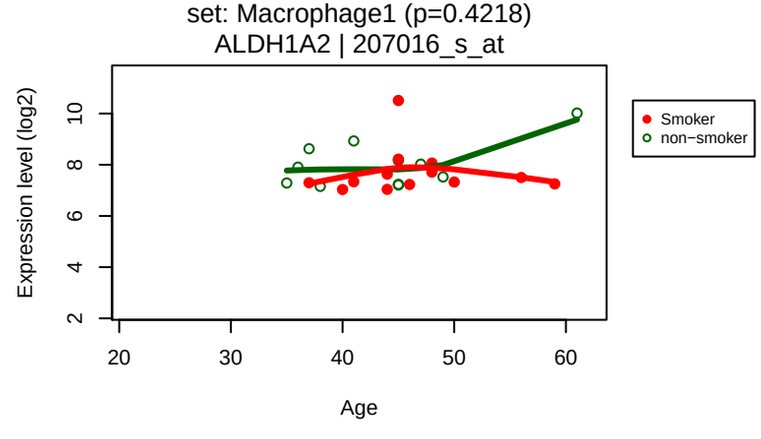

set: SmallAirway1 (p=0.3706)
ALDH1A2 | 207016_s_at
Expression level (log2)
Age
Smoker
non−smoker

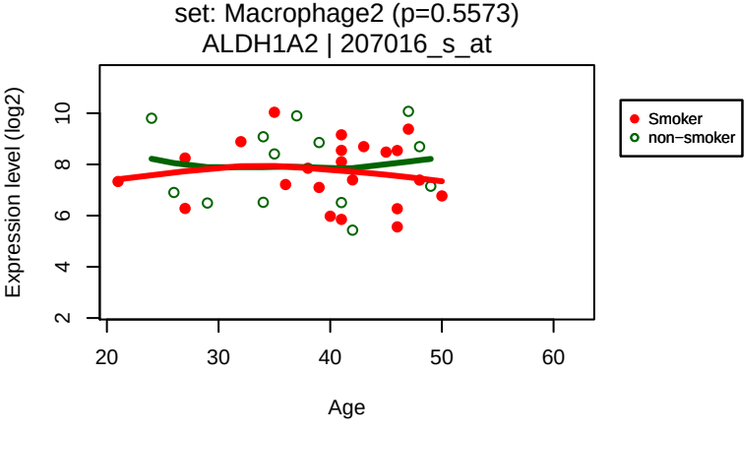

set: SmallAirway2 (p=0.2959)
ALDH1A2 | 207016_s_at
Expression level (log2)
Age
Smoker
non−smoker

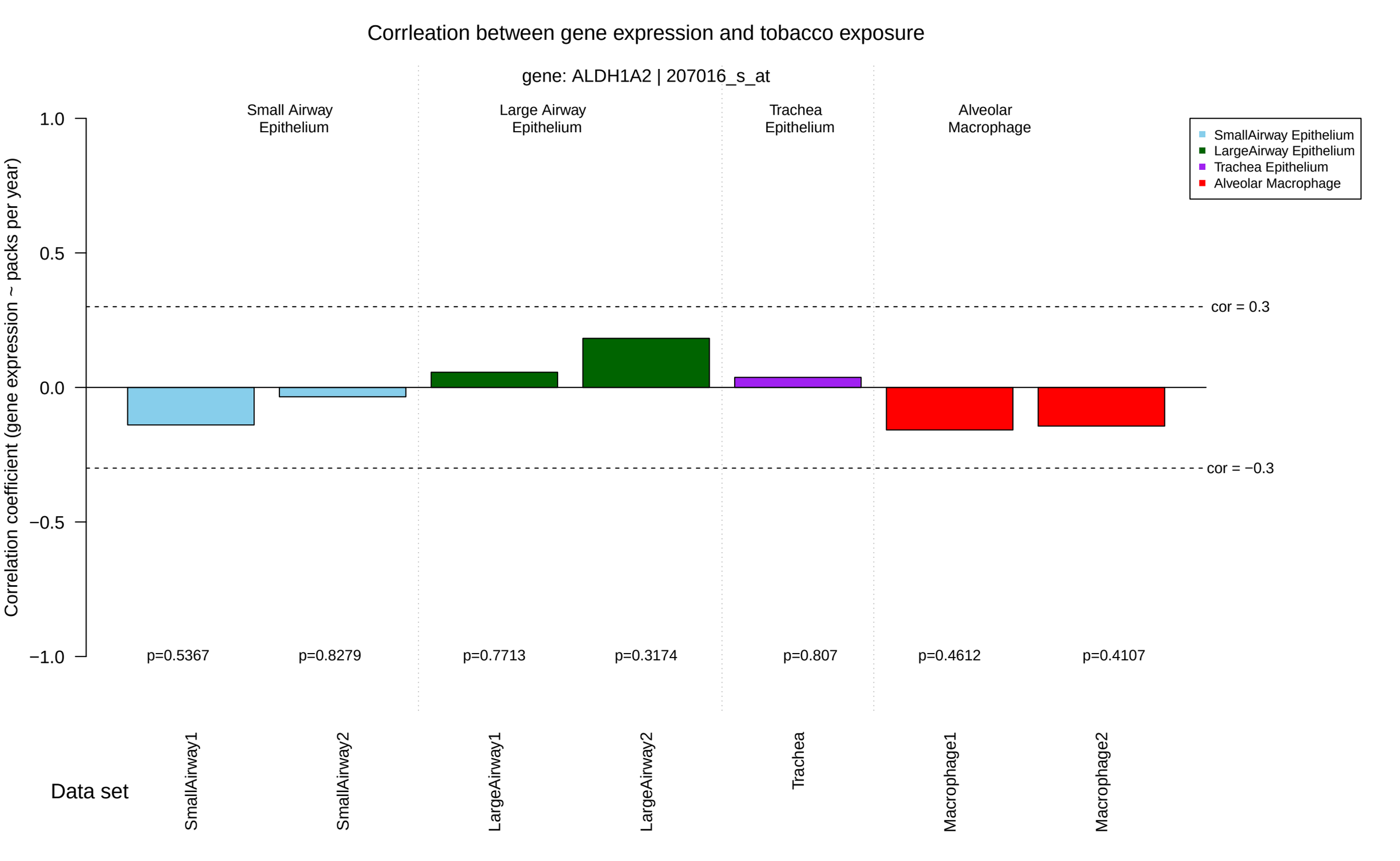

set: LargeAirway1 (p=0.6722)
ALDH1A2 | 207016_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1543)
ALDH1A2 | 207016_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.854)
ALDH1A2 | 207016_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4218)
ALDH1A2 | 207016_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5573)
ALDH1A2 | 207016_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALDH1A2 | 207016_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5367
p=0.8279
p=0.7713
p=0.3174
p=0.807
p=0.4612
p=0.4107
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2